

Question 40: Evaluate Quick Sort using hybrid approaches by combining insertion sort for small subarrays. Record execution time for different threshold values. Analyze performance improvements. Interpret the results and justify the effectiveness of hybrid sorting techniques.

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main.py  [Icons] [Run] [Share] [Clear]

1 import random
2 import time
3
4 def insertion_sort(a, low, high):
5     for i in range(low + 1, high + 1):
6         key = a[i]
7         j = i - 1
8         while j >= low and a[j] > key:
9             a[j + 1] = a[j]
10            j -= 1
11            a[j + 1] = key
12
13 def partition(a, low, high):
14     pivot = a[high]
15     i = low - 1
16
17     for j in range(low, high):
18         if a[j] <= pivot:
19             i += 1
20             a[i], a[j] = a[j], a[i]
21
22     a[i + 1], a[high] = a[high], a[i + 1]
23     return i + 1
24
25 def hybrid_quick_sort(a, low, high, threshold):
26     while low < high:
```

```
Output  [Clear]

Threshold: 5
Time: 0.023756 seconds
Threshold: 10
Time: 0.022429 seconds
Threshold: 20
Time: 0.023723 seconds
Threshold: 30
Time: 0.026769 seconds
Threshold: 50
Time: 0.020104 seconds

=== Code Execution Successful ===
```

```
main.py  [Icons] [Run] [Share] [Clear]

26 while low < high:
27     if high - low + 1 <= threshold:
28         insertion_sort(a, low, high)
29         return
30
31     p = partition(a, low, high)
32
33     if p - low < high - p:
34         hybrid_quick_sort(a, low, p - 1, threshold)
35         low = p + 1
36     else:
37         hybrid_quick_sort(a, p + 1, high, threshold)
38         high = p - 1
39
40 n = 10000
41 original = [random.randint(1, 100000) for _ in range(n)]
42
43 for threshold in [5, 10, 20, 30, 50]:
44     a = original.copy()
45
46     start = time.perf_counter()
47     hybrid_quick_sort(a, 0, len(a) - 1, threshold)
48     end = time.perf_counter()
49
50     print("Threshold:", threshold)
51     print("Time:", round(end - start, 6), "seconds")
```

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